ATOMIC STRUCTURE

PROTONS, NEUTRONS & ELECTRONS

What is an Atom?

Everything around us is made of tiny particles called atoms. They are the building blocks of all matter – from the air we breathe to the water we drink to the food we eat!

Parts of an Atom

Protons

Positive charge! Protons live in the nucleus and decide what element the atom is.

Neutrons

No charge! Neutrons are neutral and help keep the nucleus together.

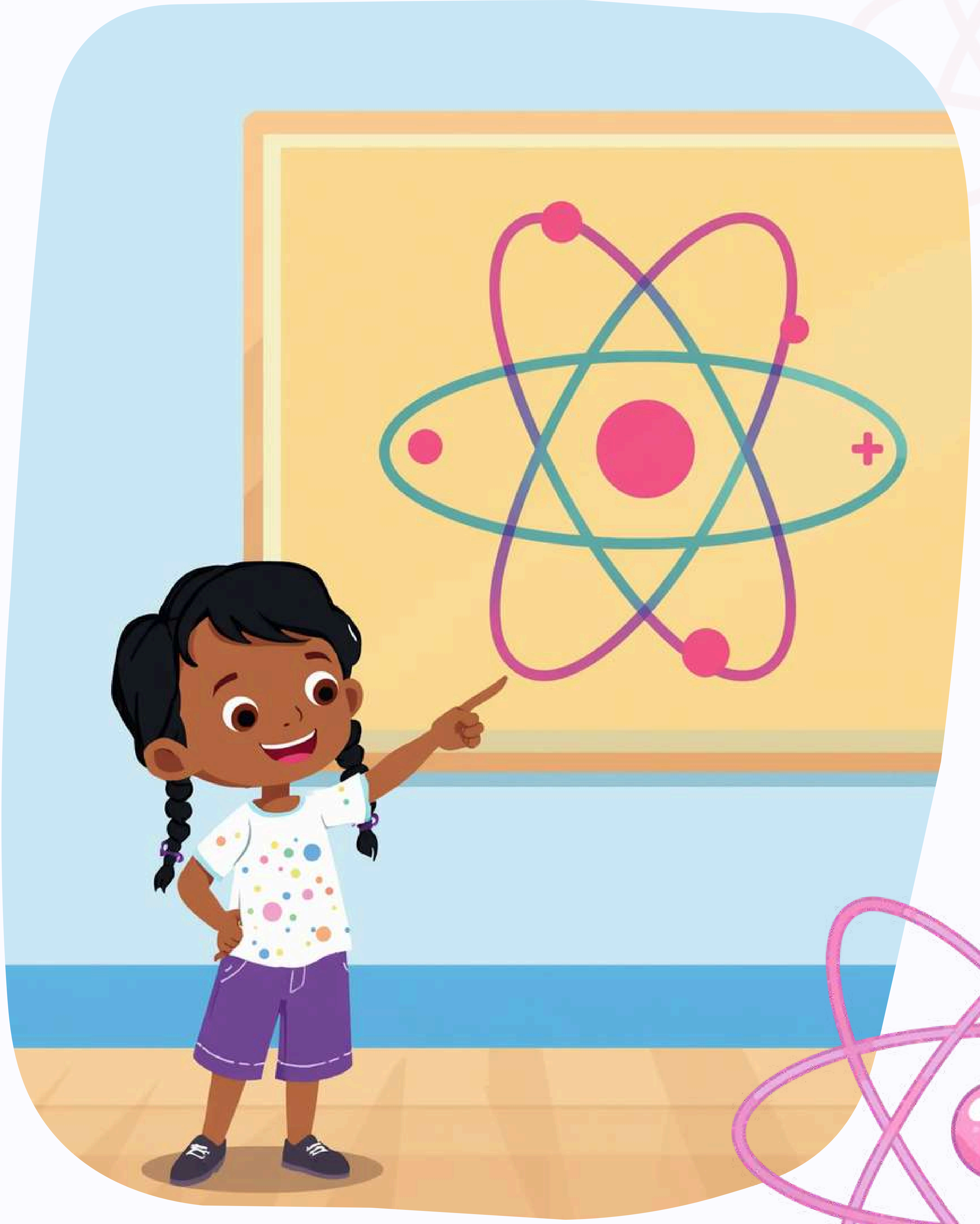
Electrons

Negative charge! Electrons zip around the nucleus super fast!

Protons: The Positive Particles

Protons are tiny particles that live inside the nucleus, which is the center of an atom. They carry a positive charge, shown as a '+' sign. Every proton is super important because it helps define what kind of element an atom is!

- Protons decide what element the atom is!
- The number of protons is called the atomic number.



Neutrons: The Neutral Particles

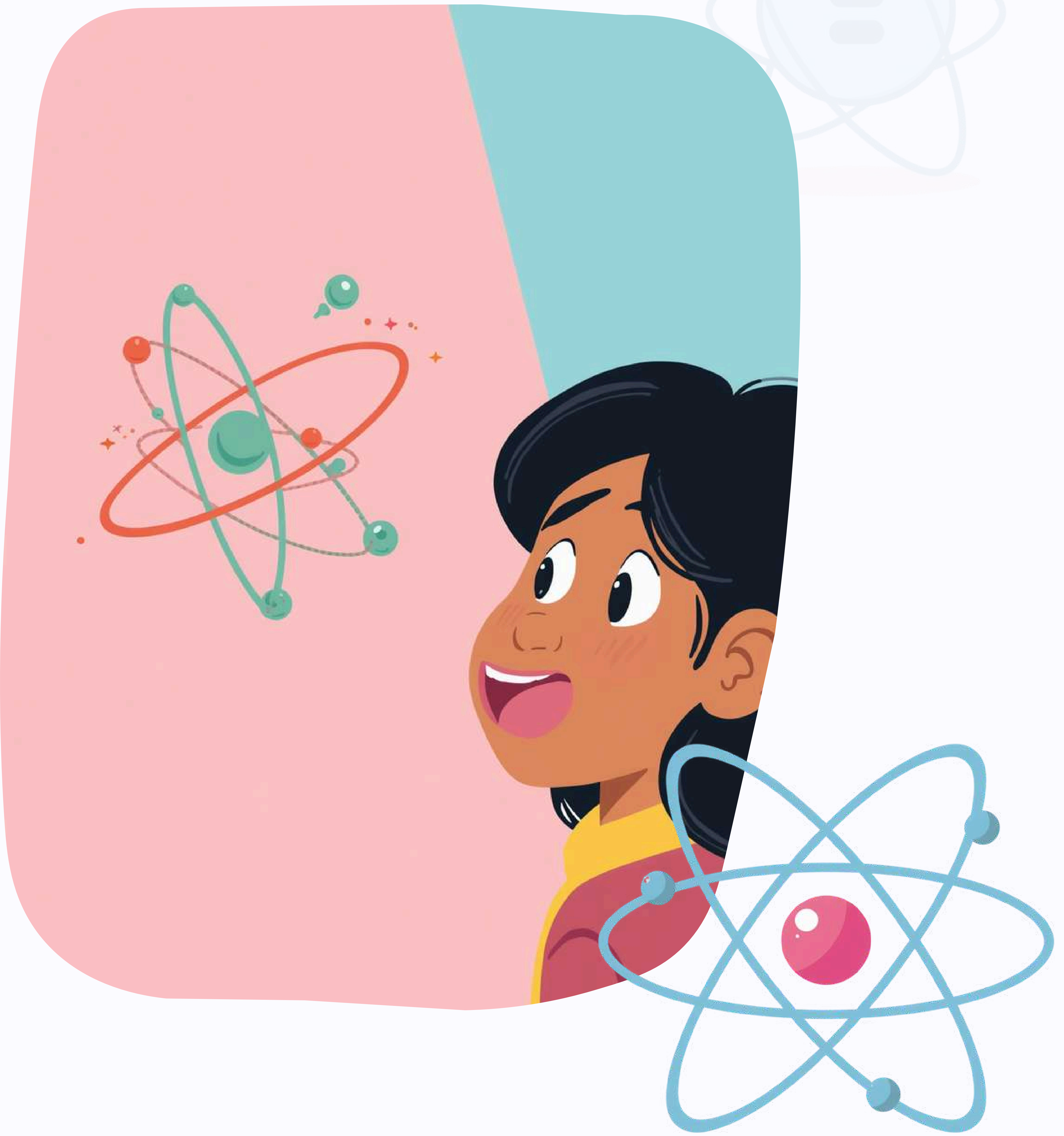
Neutrons are tiny particles that live inside the nucleus of an atom, right next to the protons. Unlike protons and electrons, neutrons have no electric charge at all – that's why we call them "neutral"!

- Neutrons help keep the nucleus together and stable.
- Without neutrons, atoms would fall apart!

Electrons: The Negative Particles

Electrons are tiny particles that orbit around the nucleus of an atom. They have a negative charge, which is the opposite of protons. Electrons are incredibly light and move in special paths called electron shells or energy levels around the nucleus.

- Electrons zip around the nucleus super fast!
- The negative charge of electrons balances the positive charge of protons.

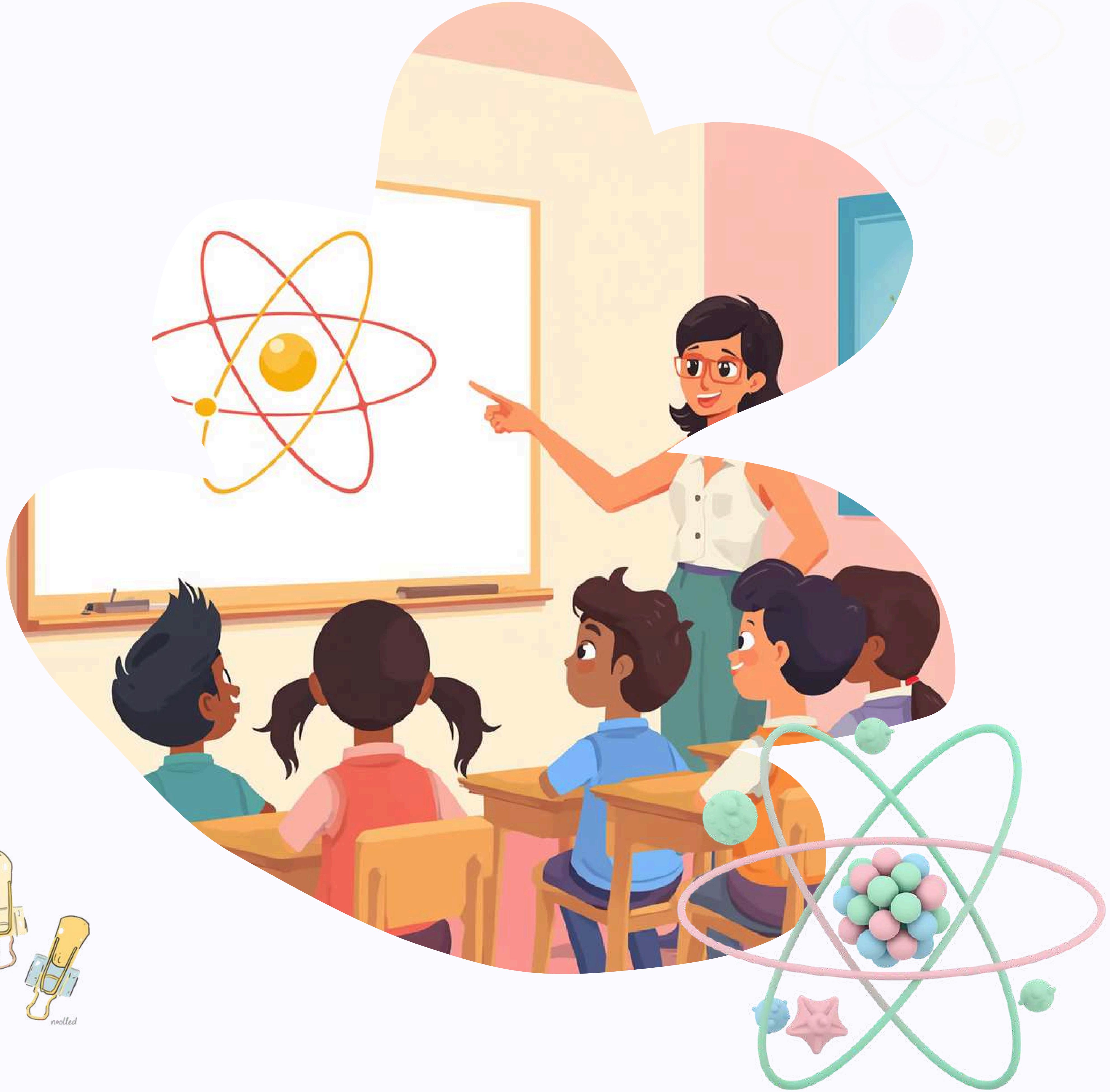


How Are They Arranged?

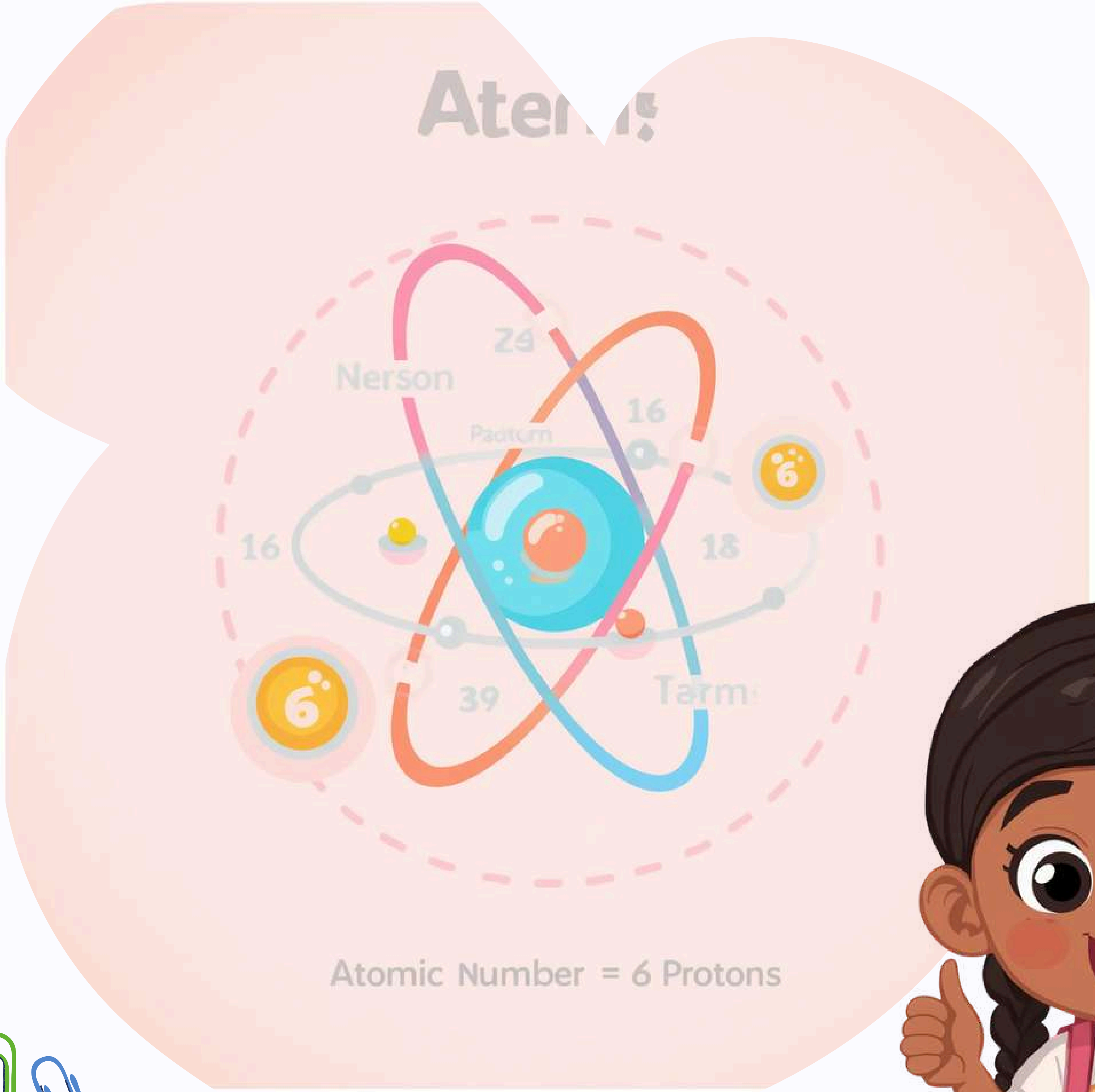
The nucleus is at the center of the atom. It contains protons (positive) and neutrons (neutral) packed tightly together.

Electrons (negative) orbit around the nucleus in special paths called electron shells. The shells are like invisible rings around the nucleus.

Think of it like our solar system – the nucleus is the sun, and electrons are planets zooming around it!



Atomic Number & Mass Number



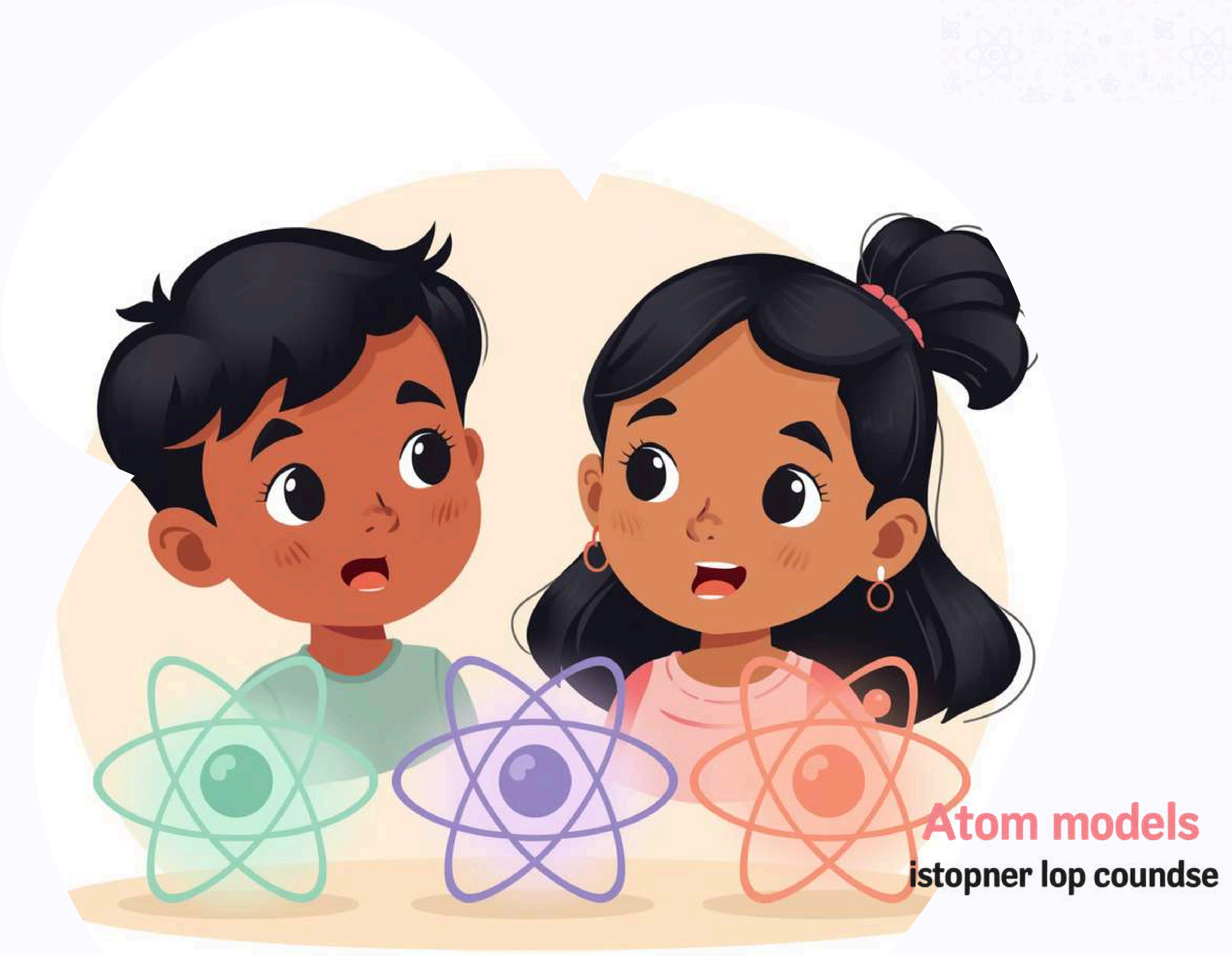
Atomic Number = Number of Protons
This tells us what element the atom is! For example, all hydrogen atoms have 1 proton.

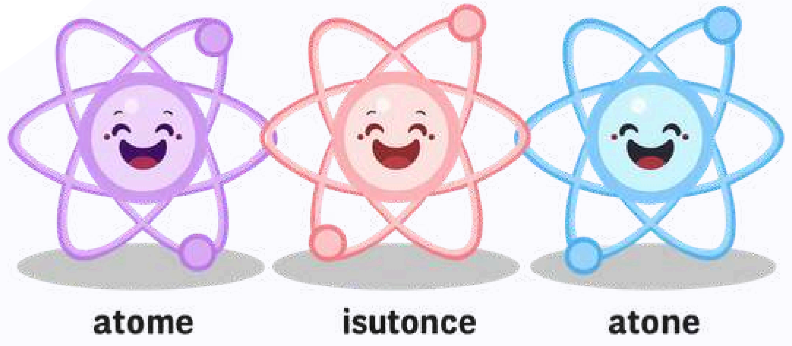
Mass Number = Protons + Neutrons
This tells us how heavy the atom is! A carbon atom has 6 protons + 6 neutrons = mass number of 12.

What Are Isotopes?

Atoms of the same element can have different numbers of neutrons. These special atoms are called isotopes!

For example, all carbon atoms have 6 protons, but some have 6 neutrons and others have 7 or 8. They're still carbon, just different versions!





Fun Facts About Atoms!

Mostly Empty!

Atoms are mostly empty space! If an atom was a stadium, the nucleus would be a tiny pea in the center.

Super Tiny!

If an apple was the size of an atom's nucleus, the atom would be as big as a football field!

Lightning Fast!

Electrons move so fast around the nucleus that we can never see exactly where they are!

Why Do Atoms Matter?

Atoms make up everything around us – from the air we breathe to the food we eat and the water we drink!

Every plant, animal, rock, and even YOU are made of tiny atoms working together. Understanding atoms helps us explore and understand our amazing world!

Recap:
What Did We Learn?

Particle Types

Protons are positive, neutrons are neutral, electrons are negative!

Atom Structure

Protons and neutrons live in the nucleus. Electrons orbit outside!

Key Numbers

Atomic number = protons. Mass number = protons + neutrons!

Quick Quiz Time!

Question 1

Which particle has a positive charge?

Question 2

Where do electrons live in an atom?

Question 3

What does the atomic number tell us about an element?

Summary
and Takeaway

Atoms are tiny but powerful building blocks of everything around us. Understanding them helps us explore and discover the amazing world of science!

Remember: Every object you see, touch, or use is made of atoms working together. From the water you drink to the air you breathe – atoms are everywhere!

Keep asking questions and stay curious about the world of atoms!

Additional Resources to Explore

Keep learning about atoms and STEM! Check out BrainPOP, National Geographic Kids, and Khan Academy for fun videos and activities.

Visit your school library for books on chemistry and physics. Ask your teacher for more exciting experiments to try at home!

THANK YOU!

Thank you for learning about atoms with us!

Questions? Keep exploring!